

1. Problem Statement

A mid-sized technology company seeks to improve employee retention by identifying key factors that influence whether employees stay. The goal is to develop a predictive model using logistic regression to classify employees based on their likelihood of remaining with the company, enabling HR to take proactive measures.

2. Methodology and Techniques Used

Data Preprocessing

- Loaded a dataset containing demographic details, job satisfaction, performance metrics, and tenure.
- Cleaned the data by handling missing values and encoding categorical variables.
- Split the dataset into training and testing sets.

Exploratory Data Analysis (EDA)

- Used plots such as histograms, countplots, and correlation heatmaps to explore patterns.
- Identified key features such as satisfaction level, number of projects, and average monthly hours as important predictors.

Modeling

- Logistic Regression was selected due to the binary classification nature of the problem.
- Fitted the model using the training data and evaluated performance using accuracy, precision, recall, and ROC-AUC score.



Key Visualizations

1. Satisfaction Level vs Retention

Insight: Employees with low satisfaction levels are significantly more likely to leave. This is a critical factor for HR to address.

2. Number of Projects vs Retention

Insight: Both underworked and overworked employees (very low or very high project count) tend to leave more often. Maintaining an optimal workload is crucial.

3. Time Spent at Company

Insight: Employees who stay for a very short time or an unusually long time have higher turnover rates. Mid-tenure employees show the highest retention.

4. Correlation Heatmap

Insight: Satisfaction level has a strong negative correlation with turnover, while evaluation score and number of projects are also important predictors.

5. ROC Curve of Logistic Regression Model

Insight: The ROC curve indicates that the logistic regression model performs well in distinguishing between employees who will stay and those who will leave.

4. Results

- **Accuracy:** The logistic regression model achieved a solid classification accuracy.
 - **Important Features:** Satisfaction level and number of projects had the highest impact on the retention decision.
 - **Insights:**
 - Employees with low satisfaction and extreme project loads (either too low or too high) are more likely to leave.
 - A moderate number of projects correlates positively with retention.
 - Time spent in the company also plays a role, with mid-tenure employees being most stable.
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5. Conclusion

This analysis shows that logistic regression is effective in predicting employee retention. HR can leverage these insights to:

- Improve employee engagement.
- Balance workloads.
- Focus on at-risk groups for retention initiatives.